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SYSTEM AND METHOD FOR AN AGRICULTURAL APPLICATOR

Abstract

A system for a boom assembly may include a retainer operably coupled with a mast frame. The retainer may define a channel extending along a rotational axis. An axle may be positioned at least partially within the retainer and may extend rearward of the base plate, the axle extending along the rotational axis. A boom assembly may include a center section configured to support one or more boom arms. The center section may include a base plate operably coupled with the center section. A bearing assembly may include a hub operably coupled with the base plate and a race assembly operably coupled with the axle. The bearing assembly may be configured to guide the rotation of the center section relative to the mast about the rotational axis.

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Background/Summary

FIELD

[0001] The present disclosure generally relates to agricultural implements and, more particularly, to systems and methods for a boom system.

BACKGROUND

[0002] Various types of vehicles utilize applicators (e.g., vehicles, floaters, etc.) to deliver an agricultural product to a ground of a field. The agricultural product may be in the form of a solution or mixture, with a carrier (such as water) being mixed with one or more active ingredients (such as an herbicide, fertilizer, fungicide, a pesticide, or another product).

[0003] The applicators may be pulled as an implement or self-propelled and can include a tank, a pump, a boom assembly, and a plurality of nozzles carried by the boom assembly at spaced locations. The boom assembly can include a pair of boom arms, with each boom arm extending to either side of the applicator when in an unfolded state. Each boom arm may include multiple boom sections, each with a number of spray nozzles (also sometimes referred to as spray tips).

[0004] During a spray operation, the vehicle drives over a target to direct the agricultural product at the target. However, the various factors may cause the boom arm to move thereby placing various sections of the boom arms at heights that are varied from a defined height above the target. Accordingly, a vehicle that is capable of altering a position of the boom assembly would be welcomed in the technology.

BRIEF DESCRIPTION

[0005] Aspects and advantages of the technology will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the technology.

[0006] In some aspects, the present subject matter is directed to a system for a boom assembly. The system includes a retainer operably coupled with a mast frame of a mast. The retainer defines a channel extending along a rotational axis. An axle is positioned at least partially within the retainer, the axle extending along the rotational axis. A boom assembly includes a center section configured to support one or more boom arms. The center section includes a base plate operably coupled with the center section, the base plate defining a void and a bearing assembly including a hub operably coupled with the base plate and a race assembly operably coupled with the axle. The bearing assembly is configured to guide rotation of the center section relative to the mast about the rotational axis.

[0007] In some aspects, the present subject matter is directed to a method of manufacturing a boom system. The method includes operably coupling a bearing assembly with a boom assembly and an axle. The bearing assembly is configured to rotate about a rotational axis and rotatable about a rotational axis. The method also includes positioning the axle within a retainer coupled with a mast, the axle extending along the rotational axis.

[0008] In some aspects, the present subject matter is directed to a system for a boom assembly that includes a retainer defining a channel extending along a rotational axis. An axle is positioned at least partially within the retainer and extends along the rotational axis. A bearing assembly is operably coupled with the axle. The bearing assembly is configured to guide rotation of a center section of a boom assembly relative to a mast about the rotational axis.

[0009] These and other features, aspects, and advantages of the present technology will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate examples of the technology and, together with the description, serve to explain the principles of the technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A full and enabling disclosure of the present technology, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0011] FIG. 1 illustrates a front perspective view of a machine in accordance with aspects of the present subject matter;

[0012] FIG. 2 illustrates a rear perspective view of the machine in accordance with aspects of the present subject matter;

[0013] FIG. 3 illustrates a side view of the machine in accordance with aspects of the present subject matter;

[0014] FIG. 4 illustrates a rear view of the machine in accordance with aspects of the present subject matter;

[0015] FIG. 5 illustrates a partial rear view of a boom system in accordance with aspects of the present subject matter;

[0016] FIG. 6 illustrates a partial bottom view of a boom system in accordance with aspects of the present subject matter;

[0017] FIG. 7 illustrates a partial side view of a boom system in accordance with aspects of the present subject matter;

[0018] FIG. 8 illustrates a cross-sectional view taken along the line VIII-VIII of FIG. 5;

[0019] FIG. 9 illustrates a perspective enhanced view of section IX of FIG. 8;

[0020] FIG. 10 illustrates a side enhanced view of section X of FIG. 8;

[0021] FIG. 11 is a schematic view of an agricultural system in accordance with aspects of the present subject matter; and

[0022] FIG. 12 illustrates a flow diagram of a method of manufacturing a boom system in accordance with aspects of the present subject matter.

[0023] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present technology.

DETAILED DESCRIPTION

[0024] Reference now will be made in detail to embodiments or examples of the disclosure, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the disclosure, not limitation of the disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the scope or spirit of the disclosure. For instance, features illustrated or described as part can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present disclosure covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0025] In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0026] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify a location or importance of the individual components. The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or

more intermediate components or features, unless otherwise specified herein. The terms “upstream” and “downstream” refer to the relative direction with respect to an agricultural product within a fluid circuit. For example, “upstream” refers to the direction from which an agricultural product flows, and “downstream” refers to the direction to which the agricultural product moves. The term “selectively” refers to a component's ability to operate in various states (e.g., an ON state and an OFF state) based on manual and/or automatic control of the component.

[0027] Furthermore, any arrangement of components to achieve the same functionality is effectively “associated” such that the functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the defined functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “operably connected” or “operably coupled” to each other to achieve the defined functionality, and any two components capable of being so associated can also be viewed as being “operably couplable” to each other to achieve the defined functionality. Some examples of operably couplable include, but are not limited to, physically mateable, physically interacting components, wirelessly interactable, wirelessly interacting components, logically interacting, and/or logically interactable components.

[0028] The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0029] Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” “generally,” and “substantially,” is not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or apparatus for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a ten percent margin.

[0030] Moreover, the technology of the present application will be described in relation to exemplary embodiments. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

[0031] As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items can be employed. For example, if a composition or assembly is described as containing components A, B, and/or C, the composition or assembly can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

[0032] In general, the present subject matter is directed to a system for a boom assembly that may include a retainer operably coupled with a mast frame. The retainer may define a channel extending along a rotational axis. An axle may be positioned at least partially within the retainer and may extend rearward of the base plate, the axle extending along the rotational axis. A boom assembly may include a center section configured to support one or more boom arms. The center section may include a base plate operably coupled with the center section. A bearing assembly may include a hub operably coupled with the base plate and a race assembly operably coupled with the axle. The bearing assembly may be configured to guide the rotation of the center section relative to the mast about the rotational axis. The bearing assembly may allow for the decoupling of the boom assembly from at least a portion of the rolling action of a chassis operably coupled with the boom assembly.

[0033] Traditionally, the rotational axis is above the range of center of gravity positions to generate a pendulum effect. This allows gravity to be a factor in locating the position to which the center

section of the boom assembly to rotate relative to a mast and/or the chassis. As such, the system for a boom assembly described herein may remove or reduce gravity's influence on a roll angle, and use the inherent inertia of the boom combined with electronics that may be hydraulically powered to influence the roll angle. Accordingly, the location of the rotational axis may be lower than in traditional boom assemblies and/or close enough to a center of gravity to minimize gravity's effect. In various instances, the rotational axis may be proximate to and/or aligned with a center of gravity of the boom assembly. Additionally, in some cases, the rotational axis may be below a center of gravity of the boom assembly. The system for a boom assembly described herein can allow for increased control, e.g., closed-loop control, of the boom assembly to determine a position of the boom assembly as the boom assembly does not have to work against gravity to the same extent as when the rotational axis is offset from the center of gravity of the boom assembly. As used herein, the rotational axis may be a point about which the center section of the boom assembly may rotate and the center of gravity may be a point at any given time where the weighted relative position of the distributed weight of the boom assembly and/or the center section and any boom arms of the boom assembly sums to zero. As such, the center of gravity may be the point to which a weight of the boom assembly is evenly distributed in all directions.

[0034] Referring now to FIGS. **1-4**, differing views of a machine **10** are illustrated in accordance with aspects of the present subject matter. In the illustrated examples, the machine **10** is configured as a self-propelled vehicle. However, in alternative examples, the machine **10** may be configured as any other suitable type of vehicle configured to perform agricultural spraying operations, such as a tractor or other vehicle configured to haul a spraying or application implement.

[0035] As shown in FIGS. **1-4**, the machine **10** may include a chassis **12** or frame configured to support or couple to a plurality of components. For example, front wheels **14** and rear wheels **16** may be coupled to the chassis **12**. The wheels **14**, **16** may be configured to support the machine **10** relative to the ground **20** and move the machine **10** in a direction of travel (e.g., as indicated by arrow **18** in FIG. **1**) across the ground **20**.

[0036] The chassis **12** may also support an operator's cab **22** that houses various control or input devices (e.g., levers, pedals, control panels, buttons, and/or the like) for permitting an operator to control the operation of the machine **10**. For instance, as shown in FIG. **1**, the machine **10** may include a human-machine or user interface **24** for displaying message windows and/or alerts to the operator and/or for allowing the operator to interface with the vehicle's controller or computing system.

[0037] Furthermore, the chassis **12** may also support one or more tanks **26**, which may be in the form of a product tank and/or an auxiliary tank. Each product tank is generally configured to store or hold an agricultural product, such as a pesticide, an herbicide, a nutrient, and/or the like. The auxiliary tank may be configured to store or hold clean water and/or any other product, which may be different from the agricultural product within the product tank.

[0038] The chassis **12** may further support a boom assembly **28** operably mounted to the chassis **12**. A plurality of nozzle assemblies **30** are mounted on the boom assembly **28** and configured to selectively dispense the agricultural product stored in the one or more tanks **26** via the nozzle assemblies **30** onto underlying plants and/or soil. The nozzle assemblies **30** are generally spaced apart from each other on the boom assembly **28** along a lateral direction **32**. Furthermore, fluid conduits may fluidly couple the nozzle assemblies **30** to the tank(s) **26**. Each nozzle assembly **30** may include a nozzle valve and an associated spray tip or spray nozzle. In several instances, the operation of each nozzle valve may be individually controlled by an associated controller or computing system such that the valve regulates the flow rate and/or another spray characteristic of the agricultural product through the associated spray nozzle.

[0039] As shown in FIGS. **1-4**, in various examples, the boom assembly **28** may include a center section **34**, a first boom arm **36**, and a second boom arm **38**. The first boom arm **36** may include a first inner boom section **36A** pivotably coupled to the center section **34**, a first middle boom section

36B pivotably coupled to the first inner boom section **36A**, and a first outer boom section **36C** pivotably coupled to the first middle boom section **36B**. Similarly, the second boom arm **38** may include a second inner boom section **38A** pivotably coupled to the center section **34**, a second middle boom section **38B** pivotably coupled to the second inner boom section **38A**, and a second outer boom section **38C** pivotably coupled to the second middle boom section **38B**. Each of the inner boom sections **36A**, **38A** is pivotably coupled to the center section **34** at pivot joints **40**. For example, the pivotable movement of the boom sections relative to one another may allow the various boom sections to move between a fully extended or working position (e.g., as shown in FIGS. **1, 2**, and **4**), in which the boom sections are unfolded along the lateral direction **32** to allow for the performance of an agricultural spraying operation, and a transport position (FIG. **3**), in which the boom sections are folded inwardly to reduce the overall width of the boom assembly **28** along the lateral direction **32**. Although the boom assembly **28** is shown in FIGS. **1-4** as including a center section **34** and three individual boom sections coupled to each side of the center section **34**, the boom assembly **28** may generally have any suitable number of boom sections. For example, in other examples, each boom arm **36**, **38** may include four or more boom sections or less than three boom sections.

[0040] In various examples, the boom assembly **28** can also include a support system **42** that may extend between the chassis **12** of the machine **10** and the mast **48** to support the mast **48**, and to facilitate height adjustment of the mast **48** relative to the ground **20**. In this manner, the height of the boom assembly **28** relative to the ground **20** may be adjusted to accommodate various crops, soil conditions, and/or delivered products, for example. In some cases, the weight of the first boom arm **32** and/or the second boom arm **34** may be supported by the center section **34**, and the center section **34** may transfer the load to the mast **48** via a linkage assembly **50** and/or a rotational assembly **52**. The mast **48**, in turn, transfers the load to the machine chassis **12** via the support system **42**, thereby suspending the boom assembly **28** above the ground **20**. In some cases, the support system **42** can include one or more links **44** and/or one or more support actuators **46**. In some instances, the one or more links **44** may be pivotably coupled, or otherwise operably coupled, with the chassis **12** and/or the mast **48**. Similarly, the one or more actuators **46** may be pivotably coupled, or otherwise operably coupled, with the chassis **12** and/or the mast **48**.

[0041] In some examples, the boom assembly **28** may include a mast **48** coupled to the center section **34**. The mast **48** can support the boom assembly **28** relative to the machine chassis **12**. For example, the center section **34** can be coupled to the mast **48** via a linkage assembly **50** and/or a rotational assembly **52**, that, in combination or individually, may be configured to transfer a downward load (e.g., as indicated by line **54**) of the center section **34** to the mast **48**. For instance, the weight of the first boom arm **36** and/or the second boom arm **38** may be supported by the center section **34**, and the center section **34** transfers the load to the mast **48** via the linkage assembly **50** and/or the rotational assembly **52**. The mast **48**, in turn, transfers the load to the machine chassis **12**, thereby suspending the boom assembly **28** above the ground **20**. Furthermore, the center section **34** may experience rotation (e.g., as indicated by line **56**) of the center section **34** relative to the mast **48** about a rotational axis **58**, which may be parallel to the direction of travel **18**. For example, if the machine **10** tilts to one side due to variations in the terrain, the center section **34**, the first boom arm **32** and/or the second boom arm **34** may rotate about the rotational axis **58**, illustrated by rotational line **56**.

[0042] In various examples, the linkage assembly **50** can include one or more position actuators **60** that may be configured to rotate the center section **34** relative to the mast **48** about the rotational axis **58**. Additionally or alternatively, the one or more position actuators **60** may be configured to adjust the height of the boom assembly **28** relative to the mast **48**.

[0043] In various instances, the rotational assembly **52** can include a retainer **72** operably coupled with a first component of the boom assembly **28** (such as the mast **48**), a bearing assembly **64** operably coupled with a second component of the boom assembly **28** (such as the center section

34), and an axle **62** operably coupled with the retainer **72** and the bearing assembly **64**. The bearing assembly **64** may rotate about the rotational axis **58**.

[0044] In various instances, the location of the rotational axis **58** may be lower than in traditional boom assemblies and/or close enough to a center of gravity **66** to minimize and/or reduce gravity's effect. In various instances, the rotational axis **58** may be proximate to and/or aligned with the center of gravity **66** of the boom assembly. Additionally, in some cases, the rotational axis **58** may be below a center of gravity **66** of the boom assembly **28**. The system for a boom assembly **28** described herein can allow for increased control, e.g., closed-loop control, of the boom assembly **28** to determine a position of the boom assembly **28** as the boom assembly **28** does not have to work against gravity to the same extent as when the rotational axis is offset from the center of gravity of the boom assembly.

[0045] Additionally, as shown in FIGS. **1-4**, the boom assembly **28** may include any number of boom section actuators **68** that are configured to alter the position of one boom section relative to another, and/or the chassis **12** of the machine **10**. In various examples, any of the boom section actuators **68** described herein may be configured as hydraulic cylinders. However, different actuators **68** may be used in other instances. For example, any of the actuators **68** may be configured as electric actuators, pneumatic cylinders, pulley systems, and/or any other practicable device.

[0046] With further reference to FIGS. **1-4**, the machine **10** may also include a sensor system **70**. In general, the sensor system **70** may be configured to capture data indicative of one or more operating conditions or parameters associated with the performance and/or operation of the machine **10**, a system operably coupled with the machine **10**, an assembly operably coupled with the machine **10**, such as the boom assembly **28**. For example, the sensor system **70** may be configured to generate data indicative of a position of the boom assembly **28** relative to a chassis **12**. The sensor system **70** may include one or more sensors **88**, a weather station, and/or any other assembly, which may be installed on the machine **10** and/or the boom assembly **28**. For instance, in some examples, the one or more sensors **88** may be installed on the boom assembly **28** to allow operating parameters/conditions associated with the boom assembly **28** to be monitored. However, one or more sensors **88** may be installed relative to or in association with any other suitable components, features, systems, and/or sub-systems of the machine **10**. In various examples, the sensors **88** may include position sensors, flow sensors, motion sensors (e.g., accelerometers, gyroscopes, etc.), image sensors (e.g., cameras, LIDAR devices, etc.), radar sensors, ultrasonic sensors, and/or the like, depending on the operating conditions being monitored.

[0047] Referring to FIGS. **4-10**, the linkage assembly **50** and/or the rotational assembly **52** may facilitate rotation of the center section **34** relative to the mast **48** in a direction **56** (i.e., about a rotational axis **58** parallel to the direction of travel **18**). For instance, a center of gravity **66** of the boom assembly **28** may be defined within the center section **34** of the boom assembly **28**. The rotational assembly **52** may define a rotational axis **58** in a direction that is generally parallel to the direction of travel **18** of the machine **10**. As a result, the boom assembly **28** may generally rotate about rotational axis **58** through the usage of the rotational assembly **52**. In some cases, by locating the rotational axis **58** proximate to the center of gravity **66** of the boom assembly **28**, increased control, e.g., closed-loop control, of the boom assembly **28** can be achieved as the boom assembly **28** does not have to work against gravity to the same extent as when the rotational axis **58** is offset by greater distances from the center of gravity **66** of the boom assembly **28**.

[0048] Referring further to FIGS. **4-10**, the rotational assembly **52** can include a retainer **72** that is operably coupled with the mast **48** (and/or the center section **34** of the boom assembly **28**) and a bearing assembly **64** operably coupled with the center section **34** of the boom assembly **28** (and/or the mast **48**). In the illustrated example, the retainer **72** may be operably coupled with a mast frame **74**, which may include one or more supports of the mast frame **74**. In some instances, the retainer **72** may define a channel **76** that is generally aligned and/or otherwise extends along the rotational

axis **58**.

[0049] An axle **62** may be positioned at least partially within the channel **76** defined by the retainer **72**. As such, in some instances, the axle **62** may also extend along the rotational axis **58**. As illustrated, a first portion **78** of the axle **62** may extend forward of one or more supports of the mast frame **74** in a fore-aft direction and a second portion **80** of the axle **62** may extend rearward of a base plate **82** in a fore-aft direction that may be operably coupled with one or more base supports **84** and/or any other component of the center section **34**.

[0050] In some cases, the retainer **72** may define one or more retainment holes **86**. In some instances, the one or more retainment holes **86** may be generally perpendicular to an extension direction (e.g., the rotational axis **58**) of the channel **76**. Similarly, the axle **62** may define a duct **88**. In some instances, the duct **88** may be aligned with the retainment holes **86** and a pin **90** may be positioned through the one or more retainment holes **86** and the duct **88** to restrict rotational movement of the axle **62** relative to the retainer **72**.

[0051] Additionally, the rotational assembly **52** can include a base plate **82** that may be operably coupled with one or more base supports **84** and/or any other component of the center section **34**. The base plate **82** may define a void **92** therethrough. In various instances, the void **92** may be offset from a center point of the base plate **82**.

[0052] In some examples, a bearing assembly **64** can include a hub **94** and a race assembly **96**. The hub **94** may be operably coupled with the base plate **82** and the race assembly **96** may be operably coupled with the axle **62**. The race assembly **96** may be configured to allow rotational movement of a rotatable portion of the bearing assembly **64** relative to the hub **94**. Moreover, the rotatable portion may be fixed or otherwise coupled to the axle **62**. As such, once assembled, the center section **34** of the boom assembly **28** may be rotatable relative to the axle **62** and/or the mast **48**. In some cases, a cover **98** may be positioned rearward of the race assembly **96** in a fore-aft direction, and/or in any other location. In various other examples, however, the base and the bearing assembly **64** may be otherwise coupled with the mast **48** and the center section **34** in any other manner that allows for rotation of the center section **34** of the boom assembly **28** relative to the mast **48**.

[0053] With reference to FIG. **11**, a boom system **100** for altering a position of one or more components of the boom assembly **28** will be described with reference to the machine **10** described above with reference to FIGS. **1-10**. However, the disclosed system **100** may generally be utilized with agricultural machines having any other suitable machine configuration. In general, when the boom assembly **28** is in the extended position (as illustrated in FIGS. **1** and **3**), the position of various sections of the boom assembly **28** may be affected due to the movement of the various sections of the boom assembly **28**. In addition, due to the configuration of the boom assembly **28**, the actuation of one boom section actuator **68** can impact the manner in which one or more of the other boom section actuators **68** may be controlled to maintain the boom assembly **28** at the defined position relative to the ground **20**, which may be based on the ground profile and/or the position profile. As such, the machine **10** may include a system **100** that is configured to determine a position (e.g., a stroke length) for each actuator **68** based on an impact to one or more remaining boom section actuators **68** to maintain the boom assembly **28** at a defined position relative to the ground **20**. Moreover, by locating the rotational axis **58** of the rotational assembly **52** proximate to the center of gravity **66** of the boom assembly **28**, increased control, e.g., closed-loop control, of the boom assembly **28** can be achieved as the boom assembly **28** does not have to work against gravity to the same extent as when the rotational axis **58** is offset from the center of gravity **66** of the boom assembly **28** by a greater distance.

[0054] As shown in FIG. **11**, the system **100** can include a computing system **102** operably coupled with various input devices **104** and one or more boom section actuators **68** within the boom assembly **28**. In general, the computing system **102** may correspond to any suitable processor-based device(s), such as a computing device or any combination of computing devices. For

example, the computing system **102** may generally include one or more processor(s) **112** and associated memory devices **114** configured to perform a variety of computer-implemented functions (e.g., performing the methods, steps, algorithms, calculations, and the like disclosed herein). As used herein, the term “processor” refers not only to integrated circuits referred to in the art as being included in a computer, but also refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application-specific integrated circuit, and other programmable circuits. Additionally, the memory **114** may generally include memory element(s) including, but not limited to, computer-readable medium (e.g., random access memory (RAM)), computer-readable non-volatile medium (e.g., a flash memory), a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD) and/or other suitable memory elements. Such memory **114** may generally be configured to store information accessible to the processor(s) **112**, including data **116** that can be retrieved, manipulated, created, and/or stored by the processor(s) **112** and instructions **118** that can be executed by the processor(s) **112**.

[0055] In several examples, the data **116** may be stored in one or more databases. For example, the memory **114** may include an input database **120** for storing input data received from the input device(s). For example, the input device(s) may include the sensor system **70**, which may include one or more sensors **88** configured to monitor one or more conditions associated with the machine **10** and/or the operation being performed therewith (e.g., including one or more of the various sensors **88**, described above), one or more positioning device(s) **106** for generating position data associated with the location of the machine **10**, one or more user interfaces **24** for allowing operator inputs to be provided to the computing system **102** (e.g., buttons, knobs, dials, levers, joysticks, touch screens, and/or the like), one or more other internal data sources **108** associated with the machine **10** (e.g., other devices, databases, etc.), one or more external data sources **110** (e.g., a remote computing device or server), and/or any other suitable input device(s). The data received from the input device(s) may, for example, be stored within the input database **120** for subsequent processing and/or analysis. It will be appreciated that, in addition to being considered an input device(s) that allows an operator to provide inputs to the computing system **102**, the user interface **24** may also function as an output device. For example, the user interface **24** may be configured to allow the computing system **102** to provide feedback to the operator (e.g., visual feedback via a display or other presentation device, audio feedback via a speaker or other audio output device, and/or the like).

[0056] Moreover, in several examples, the memory **114** may also include a location database **122** storing location information about the machine **10** and/or information about the ground **20** being processed (e.g., a field map). Such location database **122** may, for example, correspond to a separate database or may form part of the input database **120**. As shown in FIG. **11**, the computing system **102** may be communicatively coupled to the positioning device(s) **106** installed on or within the machine **10**. For example, in some examples, the positioning device(s) **106** may be configured to determine the location of the machine **10** using a satellite navigation position system (e.g., a GPS, a Galileo positioning system, the Global Navigation satellite system (GLONASS), the BeiDou Satellite Navigation and Positioning system, and/or the like). In such an example, the location determined by the positioning device(s) **106** may be transmitted to the computing system **102** (e.g., in the form of coordinates) and subsequently stored within the location database **122** for subsequent processing and/or analysis.

[0057] Referring still to FIG. **11**, in several examples, the instructions **118** stored within the memory **114** of the computing system **102** may be executed by the processor(s) **112** to implement a data analysis module **124**. In general, the data analysis module **124** may be configured to analyze the input data (e.g., a set of input data received at a given time or within a given time period or a subset of the data, which may be determined through a pre-processing method) to determine the current one or more operating parameters or conditions of the boom assembly **28** using any

algorithm and/or data processing technique. In various examples, the data analysis module **124** may implement machine learning engine methods and algorithms that utilize one or several machine learning techniques including, for example, decision tree learning, including, for example, random forest or conditional inference trees methods, neural networks, support vector machines, clustering, and Bayesian networks. These algorithms can include computer-executable code that can be retrieved by the computing system **102** and may be used to generate subsequent instructions.

[0058] In some examples, the data analysis module **124** may receive the input data from one or more input devices **104**. In turn, the system **100** may determine a current boom profile based on boom position data and defined boom dimensions/a defined boom geometry. Additionally, the system **100** may determine relative positions at various locations along the boom assembly **28** based on boom height data. In some examples, the system **100** may further calculate a ground profile based on the boom height data. The system **100** may further determine a target boom profile based on the current ground profile. Further, the system **100** may calculate a defined position for one or more boom section actuators **68** of the boom assembly **28** (e.g., a stroke length for each cylinder) to achieve the defined boom profile. In some instances, the defined position for one or more of the boom section actuators **68** of the boom assembly **28** may be based at least partially on an effect of movement of a first actuator of the one or more boom section actuators **68** on a second actuator of the one or more boom section actuators **68** (or the remaining actuators of the one or more actuators) to maintain the boom assembly **28** at the defined position relative to the ground **20**.

[0059] Referring still to FIG. **11**, the instructions **118** stored within the memory **114** of the computing system **102** may also be executed by the processor(s) **112** to implement a control module **126**. In general, the control module **126** may be configured to adjust the operation of the machine **10** by controlling one or more components of the machine **10**. In several examples, the control module **126** may be configured to control the boom assembly **28** by transmitting respective control commands to actuate each actuator **68** to a defined position to maintain the boom assembly **28** at the defined position relative to the ground **20**.

[0060] In several examples, the computing system **102** may also automatically control the operation of the user interface **24** to provide an operator notification associated with the determined one or more operating parameters or conditions of the boom assembly **28**. For instance, in some examples, the computing system **102** may control the operation of the user interface **24** in a manner that causes data associated with the determined one or more operating parameters or conditions of the boom assembly **28** to be presented to the operator of the machine **10**, such as by presenting raw or processed data associated with the one or more operating parameters or conditions of the boom assembly **28** including numerical values, graphs, maps, and/or any other suitable visual indicators.

[0061] Moreover, as shown in FIG. **11**, the computing system **102** may also include a communications interface **128** to communicate with any of the various other system components described herein. For instance, one or more communicative links or interfaces (e.g., one or more data buses and/or wireless connections) may be provided between the communications interface **128** and the input device(s) to allow data transmitted from the input device(s) to be received by the computing system **102**. Additionally, as shown in FIG. **11**, one or more communicative links or interfaces (e.g., one or more data buses and/or wireless connections) may be provided between the communications interface **128** and the boom section actuators **68** to allow the computing system **102** to control the operation of such system components.

[0062] Referring now to FIG. **12**, a flow diagram of some examples of a method **200** of manufacturing a boom system for a boom assembly is illustrated in accordance with aspects of the present subject matter. In general, the method **200** will be described herein with reference to the machine **10** and the system **100** described above with reference to FIGS. **1-11**. However, the disclosed method **200** may generally be utilized with any suitable machine **10** and/or may be utilized in connection with a system having any other suitable system configuration. In addition, although FIG. **12** depicts steps performed in a particular order for purposes of illustration and

discussion, the methods discussed herein are not limited to any particular order or arrangement. One skilled in the art, using the disclosures provided herein, will appreciate that various steps of the methods disclosed herein can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure.

[0063] As shown in FIG. 12, at (202), the method 200 can include operably coupling a bearing assembly with a boom assembly and an axle. The bearing assembly may be configured to rotate about a rotational axis and rotatable about a rotational axis. At (204), the method 200 can include positioning the axle within a retainer coupled with a mast. In some examples, the axle may extend along the rotational axis within a channel defined by the retainer. At (206), the method 200 can include positioning a pin through one or more retainment holes defined by the retainer and a duct defined through the axle to restrict rotational movement of the axle relative to the retainer. At (208), the method 200 can include operably coupling a linkage assembly to the mast and the boom assembly. In some cases, the linkage assembly may be configured to rotate a center section of the boom assembly relative to the mast about the rotational axis.

[0064] At (210), the method 200 can include operably coupling a sensor system with the boom assembly. In various examples, the sensor system may be configured to generate data indicative of a position of the boom assembly relative to a chassis. At (212), the method 200 can include operably coupling a computing system with the linkage assembly. In various instances, the computing system may be configured to alter a stroke of one or more actuators of the linkage assembly based on the position of the boom assembly relative to the chassis.

[0065] It is to be understood that the steps of any method disclosed herein may be performed by a computing system upon loading and executing software code or instructions that are tangibly stored on a tangible computer-readable medium, such as on a magnetic medium, e.g., a computer hard drive, an optical medium, e.g., an optical disc, solid-state memory, e.g., flash memory, or other storage media known in the art. Thus, any of the functionality performed by the computing system described herein, such as any of the disclosed methods, may be implemented in software code or instructions that are tangibly stored on a tangible computer-readable medium. The computing system loads the software code or instructions via a direct interface with the computer-readable medium or via a wired and/or wireless network. Upon loading and executing such software code or instructions by the controller, the computing system may perform any of the functionality of the computing system described herein, including any steps of the disclosed methods.

[0066] The term “software code” or “code” used herein refers to any instructions or set of instructions that influence the operation of a computer or controller. They may exist in a computer-executable form, such as vehicle code, which is the set of instructions and data directly executed by a computer's central processing unit or by a controller, a human-understandable form, such as source code, which may be compiled in order to be executed by a computer's central processing unit or by a controller, or an intermediate form, such as object code, which is produced by a compiler. As used herein, the term “software code” or “code” also may include any human-understandable computer instructions or set of instructions, e.g., a script, that may be executed on the fly with the aid of an interpreter executed by a computer's central processing unit or by a controller.

[0067] This written description uses examples to disclose the technology, including the best mode, and also to enable any person skilled in the art to practice the technology, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the technology is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A system for a boom assembly, the system comprising: a retainer operably coupled with a mast frame of a mast, the retainer defining a channel extending along a rotational axis; an axle positioned at least partially within the retainer, the axle extending along the rotational axis; and a boom assembly including a center section configured to support one or more boom arms, the center section comprising: a base plate operably coupled with the center section, the base plate defining a void; and a bearing assembly including a hub operably coupled with the base plate and a race assembly operably coupled with the axle, wherein the bearing assembly is configured to guide rotation of the center section relative to the mast about the rotational axis.
2. The system of claim 1, wherein the retainer defines one or more retainment holes that are generally perpendicular to an extension direction of the channel.
3. The system of claim 2, wherein the axle defines a duct, and wherein the duct is aligned with the one or more retainment holes.
4. The system of claim 3, further comprising: a pin positioned through the one or more retainment holes and the duct to restrict rotational movement of the axle relative to the retainer.
5. The system of claim 1, wherein a first portion of the axle extends forward of one or more supports of the mast frame in a fore-aft direction and a second portion of the axle extends rearward of the base plate in the fore-aft direction.
6. The system of claim 1, further comprising: a cover positioned rearward of the race assembly in a fore-aft direction.
7. The system of claim 1, wherein the void is offset from a center point of the base plate.
8. The system of claim 1, further comprising: a linkage assembly configured to rotate the center section relative to the mast about the rotational axis.
9. The system of claim 8, wherein the linkage assembly comprises one or more actuators configured to rotate the center section relative to the mast about the rotational axis.
10. The system of claim 9, further comprising: a sensor system configured to generate data indicative of a position of the boom assembly relative to a chassis.
11. The system of claim 10, further comprising: a computing system configured to alter a stroke of the one or more actuators based on the position of the boom assembly relative to the chassis.
12. A method of manufacturing a boom system, the method comprising: operably coupling a bearing assembly with a boom assembly and an axle, the bearing assembly configured to rotate about a rotational axis and rotatable about a rotational axis; and positioning the axle within a retainer coupled with a mast, the axle extending along the rotational axis.
13. The method of claim 12, further comprising: positioning a pin through one or more retainment holes defined by the retainer and a duct defined through the axle to restrict rotational movement of the axle relative to the retainer.
14. The method of claim 12, further comprising: operably coupling a linkage assembly to the mast and the boom assembly, wherein the linkage assembly is configured to rotate a center section of the boom assembly relative to the mast about the rotational axis.
15. The method of claim 14, further comprising: operably coupling a sensor system with the boom assembly, the sensor system configured to generate data indicative of a position of the boom assembly relative to a chassis; and operably coupling a computing system with the linkage assembly, the computing system configured to alter a stroke of one or more actuators of the linkage assembly based on the position of the boom assembly relative to the chassis.
16. A system for a boom assembly, the system comprising: a retainer defining a channel extending along a rotational axis; and an axle positioned at least partially within the retainer and extending along the rotational axis; a bearing assembly operably coupled with the axle, wherein the bearing assembly is configured to guide rotation of a center section of a boom assembly relative to a mast

about the rotational axis.

17. The system of claim 16, wherein the retainer is operably coupled with the mast and the bearing assembly is operably coupled with a base plate, the base plate operably coupled with the center section of the boom assembly.

18. The system of claim 16, wherein the bearing assembly includes a hub operably coupled with a base plate and a race assembly operably coupled with the axle, and wherein the bearing assembly is configured to guide rotation of the center section relative to the mast about the rotational axis.

19. The system of claim 16, wherein the retainer defines one or more retainment holes that are generally perpendicular to an extension direction of the channel, wherein the axle defines a duct, and wherein the duct is aligned with the one or more retainment holes.

20. The system of claim 19, further comprising: a pin positioned through the one or more retainment holes and the duct to restrict rotational movement of the axle relative to the retainer.
